

PVTRACE · REQUIREMENTS

PVTrace requirements

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/pvtrace		
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REVISION HISTORY

Version	Date	Author	Change
0.3	2026-09-25	Amish Chadha	Status from PVT-CAL-001; decisions from PVT-DDR-001 (6.6 mF, isolation, grade thresholds); R10 and R17 wording; 33 J stored energy
0.2	2026-09-25	Amish Chadha	First measurable requirements with concept status for TRL 2
0.1	2026-09-25	Amish Chadha	Initial scaffold

PVTrace requirements

These requirements are proposals for review, not yet agreed with users, and will be revised after co-design (see PVT-PRB-001). The status column gives the design's position against each target from the calculations in PVT-CAL-001 (TRL 3). Four requirements are **not met** on paper (R3 at worst-case capacitor tolerance, R13 display readability, R16 and R17 at the \$150 budget), two are at risk (R7, R8) and two cannot be verified until hardware exists (R6, R9). Design choices adopted for TRL 3 work are listed in PVT-DDR-001; they are open for Amish's review.

The **reference module** is a 144 half-cell module of about 450 W: open-circuit voltage (Voc) about 49.5 V, short-circuit current (Isc) about 11.6 A. The **high-current case** is a 108 half-cell 182 mm module of about 410 W (Voc about 37.5 V, Isc about 13.9 A). The **high-voltage case** is two older 60-cell 250 W modules in series (Voc about 75 V, Isc about 8.9 A).

Table 1. Requirements

ID	Requirement	Target	Verification (TRL 3 or later)	Status (PVT-CAL-001)
R1	Voltage range	0 to 100 V DC open circuit, including cold-weather Voc rise; hard limit below 120 V	Component ratings; calculation	Met by design review: high-voltage case 83 V at -10 °C; 160 V capacitors, 150 V MOSFETs. Two 450 W modules in series (109 V cold) are refused
R2	Current range	0 to 20 A short circuit	Shunt and switch ratings	Met by design review: 20.48 A full scale; shunt 1.6 W of 3 W
R3	Sweep duration, to limit capacitive error in high-efficiency cells	20 to 200 ms from Isc to Voc for any module in range	Sweep model; later bench sweep	Not met at worst-case tolerance: high-current case 20.5 ms at 6.6 mF nominal, 16.4 ms at -20 %; longest 81 ms
R4	Curve resolution	200 or more V-I pairs per sweep	Sample-rate calculation	Met: 410 pairs or more
R5	Measurement accuracy after calibration	Voltage and current within ± 1 % of reading ± 0.1 % of full scale	Error budget; later check against a calibrated meter	Met on paper: about ± 0.2 % of reading ± 0.03 % of full scale after calibration
R6	Repeatability	Pmax within ± 1 % over 3 consecutive sweeps at stable irradiance (± 1 % change)	Later field test	Not verifiable at TRL 3; instrument noise about 0.08 % per sample
R7	Irradiance and temperature	Plane-of-array irradiance within ± 5 % from a reference cell; module back temperature within ± 2 °C	Calibration plan; CalRig for the temperature probe	At risk: needs a reference cell calibrated within ± 4.6 %
R8	STC translation	Pmax translated to STC per IEC 60891 with an expanded uncertainty of ± 5 % or better at 700 W/m ² or more	Uncertainty budget	At risk: ± 3.7 to ± 5.4 %; met if the reference cell is within ± 4.5 %
R9	Fault flags	Flag steps from bypass diode conduction (loss of one substring or more), raised series resistance, lowered shunt resistance and an Isc deficit against irradiance; shown within 5 s of the sweep	Rule definitions; later tests on modules with known faults	Not verifiable at TRL 3; about 7 or more samples per volt resolve a substring step
R10	Second-life grading	Grade A, B, C or reject from STC Pmax against nameplate, with thresholds held in an editable table that a partner can change	Rule review with partner	Met by design review; thresholds adopted for TRL 3 (PVT-DDR-001 item 8), still to be agreed with a partner
R11	Safe connection and disconnection	Reverse polarity and over-voltage detected before the load connects; the circuit is never opened above 0.5 A; load capacitors below 30 V within 2 s after each sweep and below 60 V within 60 s if the controller fails	Circuit review; discharge calculation	Met: 0.21 s to 30 V; 40 s passive to 60 V; load switch held up to 6 ms past 99 % of Voc so it opens below 0.5 A

ID	Requirement	Target	Verification (TRL 3 or later)	Status (PVT-CAL-001)
R12	Handheld	Mass 1.5 kg or less including leads and sensor pod; enclosure 250 x 150 x 100 mm or less	Mass estimate; model	Met: 1.41 kg; 234 x 144 x 80 mm over bumpers, 98 mm to the knob top
R13	Field conditions	Operate at 0 to 45 °C ambient in full sun; IP54 when closed; display readable in direct sun	Datasheets; heat model; later field check	IP54 met by design; display readability not met with the adopted TFT; heat at risk (about 56 °C inside a light case at 45 °C in sun)
R14	Battery life	8 h of field use or 200 sweeps per charge; USB-C charging	Power budget	Met: 10.3 h
R15	Open data	Every sweep saved as CSV (raw V-I pairs, irradiance, temperature, time, module ID, flags, grade); export by USB or Wi-Fi to a phone	Design review (no firmware at TRL 3)	Met by design review
R16	Second-life safety screening	Detect insulation faults (cracked backsheet, wet leakage) before a module is resold	Not in this instrument	Not met: out of scope; an insulation tester is needed alongside PVTrace
R17	Cost	Parts for one prototype within budget_usd (\$150; \$165 proposed, awaiting Amish)	Priced BOM (bom/bom.csv)	Not met at \$150: \$163; met at the proposed \$165 with \$2 margin

■ Assumptions

- Module curves are modeled with a single-diode model fitted to datasheet Voc, Isc, Vmp and Imp; sweep times are the time for a 6.6 mF load ($\pm 20\%$) to reach 99 % of Voc, with 56.6 m Ω loop resistance (PVT-CAL-001).
- The ADC delivers at least 25,000 V-I pairs per second in practice (half the rated rate of a 100 kS/s two-channel converter, to allow for SPI and firmware overhead).
- The reference cell is calibrated once against a pyranometer or a module of known power under clear sky; its uncertainty (about ± 3 to 5 %) dominates R8.
- The 20 ms lower bound in R3 is a working figure for high-efficiency (PERC, TOPCon, HJT) cells, whose capacitance distorts fast sweeps; it could not be checked against literature at TRL 3 and stays a working figure. PVT-CAL-001 shows the capacitive current error is about $C_{\text{module}} / C_{\text{load}}$, so it depends on the load capacitance more than on the sweep time.
- 700 W/m² follows IEC 61829 guidance for converting to STC ([Seaward](#)).

Safety: The tracer connects directly to energized PV modules at up to 100 V DC and 20 A, and stores up to 33 J in its load capacitors. R11 exists to keep the user safe; it must not be relaxed to meet cost.